

NEW DELHI SAFDARJUNG, India

WMO: 421820

Lat: 28.5886N

Lon: 77.2222E

Elev: 705

StdP: 14.33

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.8	44.7	33.0	28.2	69.7	36.2	32.2	67.4	16.2	66.8	14.2	65.1	1.1	270	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.4	108.2	73.8	105.7	74.5	103.3	74.6	84.1	94.3	83.4	93.1	82.7	92.0	8.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.9	170.4	88.4	81.1	166.2	87.9	80.6	163.2	87.6	49.2	94.4	48.4	93.4	47.6	92.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	13.2	11.2	DB	39.4	111.8	2.6	1.8	37.5	113.1	36.0	114.2	34.5	115.2	32.6	116.4	
			WB	38.6	85.6	2.1	1.1	37.1	86.4	35.8	87.0	34.6	87.6	33.1	88.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	57.3	64.1	74.1	85.0	91.3	92.0	88.5	86.7	85.3	79.5	69.5	60.3
	DBStd	12.65	4.15	4.80	5.56	5.00	4.86	5.25	3.80	2.91	3.13	4.01	3.91	4.47
	HDD50	3	2	0	0	0	0	0	0	0	0	0	0	1
	HDD65	475	240	69	3	0	0	0	0	0	0	0	7	156
	CDD50	10172	229	394	746	1051	1280	1260	1193	1139	1060	915	585	320
	CDD65	5169	2	43	284	601	815	810	728	674	610	450	142	10
	CDH74	69473	24	373	2825	8122	12728	12755	10357	9016	7644	4411	1117	101
	CDH80	39975	0	68	1251	4900	8512	8525	5947	4635	3745	2119	269	4
(14) Wind	WSAvg	3.5	3.1	3.8	4.3	4.4	4.7	4.8	4.2	3.9	3.4	2.0	1.7	2.2
(15) Precipitation	PrecAvg	30.7	0.7	0.9	0.7	0.4	0.9	3.1	8.3	9.7	5.2	0.7	0.2	0.4
	PrecMax	48.5	3.1	4.8	3.8	2.4	3.1	8.0	24.9	23.0	14.9	3.7	1.5	2.8
	PrecMin	15.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	2.1	0.0	0.0	0.0	0.0
	PrecStd	9.1	0.7	1.0	0.8	0.6	0.9	2.0	5.4	5.2	3.6	1.0	0.3	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	86.3	98.7	107.0	111.4	111.5	103.9	98.6	98.1	96.5	88.8	80.0
		MCWB	62.5	66.4	68.6	71.0	72.0	73.7	79.5	81.0	77.7	74.5	67.3	62.8
	2%	DB	73.2	81.7	93.9	104.0	108.7	108.5	100.6	96.6	96.2	94.2	85.6	76.8
		MCWB	59.9	64.5	67.9	70.6	72.7	75.3	80.5	81.3	77.9	74.1	66.9	62.0
	5%	DB	70.2	78.5	90.4	101.6	106.2	105.8	98.2	95.0	94.8	92.4	83.2	74.2
		MCWB	57.5	62.2	67.0	69.9	73.1	77.1	81.1	81.4	78.1	73.4	65.8	60.2
	10%	DB	67.4	75.5	87.2	98.8	103.8	102.9	96.0	93.3	93.1	90.2	80.7	71.5
		MCWB	56.1	60.7	66.3	69.3	73.5	77.8	81.6	81.4	78.2	72.4	64.7	58.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	69.4	73.7	78.1	82.8	84.5	85.1	84.6	83.6	80.7	71.9	66.4
		MCDB	73.7	79.5	89.6	99.1	100.3	97.9	93.8	92.8	92.1	90.9	82.4	75.7
	2%	WB	61.9	66.5	71.3	75.2	80.6	83.5	84.2	83.7	82.4	78.6	69.9	63.8
		MCDB	70.0	78.5	87.0	95.0	97.7	97.4	93.7	91.7	90.5	88.4	81.1	73.3
	5%	WB	59.8	64.4	69.6	73.2	78.9	82.6	83.5	83.0	81.4	76.9	68.3	62.0
		MCDB	66.8	75.1	84.2	93.5	96.6	95.7	92.6	90.8	89.2	85.9	78.7	71.1
	10%	WB	58.0	62.6	68.0	71.7	77.3	81.7	82.8	82.4	80.5	75.4	66.7	60.1
		MCDB	64.4	71.5	82.0	92.3	95.4	94.1	91.5	90.0	88.0	85.1	76.8	68.8
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	21.7	23.4	25.0	22.7	17.4	12.3	11.4	14.3	21.5	24.4	22.7
		MCDBR	24.2	24.6	26.7	27.5	25.1	20.8	15.0	13.7	16.6	22.4	25.8	26.0
	5% WB	MCWBR	11.5	9.5	7.5	7.9	7.5	6.8	5.0	4.9	4.9	6.8	9.9	12.2
		MCWBR	20.9	22.5	24.4	24.2	21.1	16.2	12.8	12.0	13.4	17.8	22.9	23.7
(40) Clear-Sky Solar Irradiance	taub		0.511	0.487	0.502	0.577	0.675	0.745	0.751	0.656	0.584	0.662	0.636	0.574
		taud	1.814	1.931	1.924	1.784	1.634	1.551	1.563	1.755	1.872	1.620	1.619	1.695
	Ebn at Noon		221	245	251	237	214	199	197	215	226	193	184	193
		Edh at Noon	58	56	59	70	82	89	88	72	62	75	70	63
(44) All-Sky Solar Radiation	RadAvg		958	1357	1784	2095	2145	1894	1559	1532	1593	1458	1132	961
	RadStd		118	112	83	66	96	143	136	132	143	71	74	72

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.52	N/A	+3.62	N/A	+0.68	+0.59	N/A	N/A	+184
(47) Regional (2 neighbors)		N/A	N/A	+1.89	N/A	N/A	N/A	N/A	N/A	+193

Nomenclature: See separate page